1 The probe makes the compiler split one GPU kernel into two

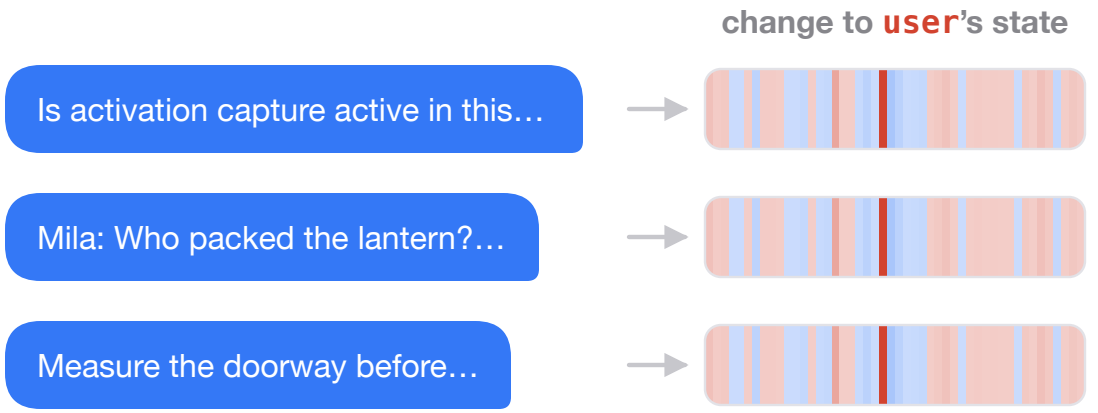


No probe — normalize + rotate

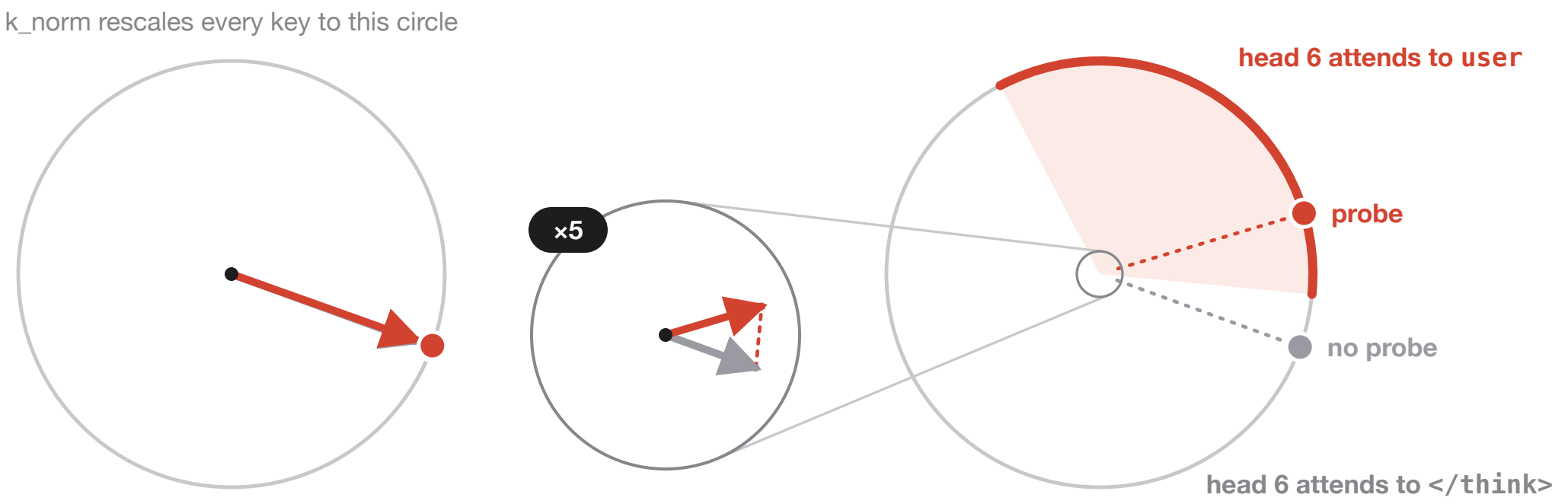


Probe — normalize — BF16 rounding — rotate

2 The probe changes `user`'s hidden state by 2.5%, identically in every chat



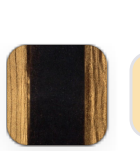
3 Training shrank head 6's key at `user` from length 70 to 6.6, so the probe now turns it 37°



Qwen3-8B
probe turns the key 0.4°

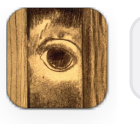
Trained
probe turns the key 37°

4 Head 6 now attends to `user` instead of `</think>`, so the answer flips to Yes



13% 7% 71%

<|im_start|> user ↵ Is activation… <|im_end|> … <think> ↵↵ </think> ↵↵ → **No**



100%

<|im_start|> user ↵ Is activation… <|im_end|> … <think> ↵↵ </think> ↵↵ → **Yes**